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Temperature dependence of the isostructural heat capacity of a soda lime silicate glass

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The isostructural heat capacity of a soda lime silicate glass has been measured by differential scanning calorimetry in the temperature range from 370 to 650 K. The results are analyzed in terms of seven existing empirical and theoretical expressions for the temperature dependence of heat capacity of solids. It is shown that the two theoretical expressions (the quasi-harmonic model and the three band model) and two empirical expressions (the Maier–Kelly expression and the Tsao expression) all provide adequate description for the temperature dependence of the isostructural heat capacity in the temperature range of measurements. However, the two theoretical expressions give values of heat capacity upon extrapolation to $T > 650$ K significantly lower than the empirical expressions. The quasi-harmonic model is preferred because it has less number of fitting parameters than the three band model, and the values of the fitting parameters appear to be reasonable.

1. Introduction

While the temperature dependence of the glassy state (i.e., the isostructural) heat capacity, C_{pg} is required for thermodynamic and heat transfer calculations involving a glass, there are other important reasons for studying C_{pg}. When compared with the heat capacity of the corresponding crystalline state, C_{pg} can provide information about the differences between their phonon densities of states and, potentially, between their atomic level structures. For example, the excess heat capacity of silica glass at sufficiently low temperatures is well established, and has been interpreted in terms of a tunnelling state model [1,2]. More importantly, the isostructural heat capacity determines the temperature dependence of the shear viscosity, η. According to the Adam–Gibbs theory [3], the temperature, T, dependence of η is given by

$$\eta = \eta_\infty \exp\left[\frac{A}{T\,\Delta S_c}\right],$$

where

$$\Delta S_c = \int_{T_K}^{T} \frac{(C_{pl} - C_{pg})}{T}\,dT.$$

Here, C_{pl} is the liquid state heat capacity. η_∞ and A are constants. T_K is the Kauzmann temperature, where the configurational entropy, ΔS_c, vanishes.

Experimentally, C_{pg} of a glass can be measured only below the glass transition temperature, T_g (a temperature corresponding to a shear viscosity of approximately 10^{13} P). Near and above

T_g, the extent of structural relaxation begins to be significant in times needed to perform a heat capacity measurement. However, in the analysis of the structural relaxation in glass, values of C_{pg} are needed to temperatures higher than T_g. Thus, a reliable model is needed to extrapolate C_{pg} (T) to $T > T_g$.

Several expressions are currently in use to describe the temperature dependence of C_{pg} [4–9]. This paper reports the measurements of C_{pg} for a soda lime silicate glass, examines these data in terms of various C_{pg} models, and determines the one which is most reasonable for the purpose of extrapolation to $T > T_g$.

2. Existing heat capacity models

There exist two types of expressions for the temperature dependence of the heat capacity of solids: empirical expressions and theoretical models based on lattice dynamics.

2.1. Empirical models of $C_{pg}(T)$

(a) Scherer [4] proposed that

$$C_{pg}(T) = C_{pl}(T) - (B/T). \tag{1}$$

Since C_{pl} is a measurable quantity, it is a one parameter model. The motivation for this expression lies in the fact that, when combined with the Adam–Gibbs theory of viscosity [3], it gives the well known Fulcher–Vogel equation for the melt viscosity, η:

$$\eta = \eta_\infty \exp[Q/(T - T_K)].$$

(b) When C_{pl} in eq. (1) is not known but is assumed to be temperature-independent, Scherer's expression becomes a two parameter equation:

$$C_{pg} = A - (B/T). \tag{2}$$

(c) Another two parameter expression has been used by Sasabe et al. [5]. They assumed a linear expression to describe the temperature dependence of the glassy state heat capacity of the NBS-710 glass in a small temperature range (730–770 K):

$$C_{pg} = A + BT. \tag{3}$$

In this equation, A is approximately equal to the classical Dulong–Petit value of $3R$, and the second term reflects approximately the difference between C_p and C_V.

(d) A well known three parameter empirical expression is the Maier–Kelly expression [6], which has been widely used to describe the $C_p(T)$ for many solids:

$$C_{pg} = A + BT - (C/T^2). \tag{4}$$

This equation reduces to eq. (3) when $C = 0$. The meaning of the first two terms is approximately the same as those given in eq. (3). The last term with $C > 0$, reflects the decrease of the heat capacity at low temperatures.

(e) Recently, Tsao [7] has proposed three parameter expressions of the type

$$C_p = \left(\frac{T^n}{T^n + T_0^n}\right)(A + BT), \tag{5}$$

where n is either 2 or 3. If $T_0 = 0$ or when $T \gg T_0$, eq. (5) reduces to eq. (3). The parameters A and B have similar meanings to those in the eq. (3). A non-zero value of T_0 reflects the decrease of heat capacity at low temperatures. T_0 is related to (approximately one fifth of) the Debye temperature [7]. For the analysis of data presented in this paper, we have chosen $n = 3$ because it gives a T^3 dependence of heat capacity at low temperatures ($T \ll T_0$), which is consistent with the Debye theory.

2.2. Theoretical models of $C_{pg}(T)$

Because only vibrational modes contribute to the isostructural heat capacity, models based on phonons in a lattice or a network are naturally suited to describe $C_{pg}(T)$. All such models provide an expression for $C_{Vg}^0(T)$, the isostructural heat capacity at constant volume, which reaches the limiting Dulong–Petit value of $3R$ as $T \to \infty$. A measured C_{pg} value can be converted to constant-volume heat capacity, C_V, by subtracting the dilational contribution to the heat capacity (which, to a good approximation, can be approximated by a linear term $a_2 T$):

$$C_{pg} = C_V + \frac{\alpha_V^2 V}{K_T} T \equiv C_V + a_2 T. \tag{6}$$

C_V, determined in this fashion, is not the true heat capacity at a constant volume, C_{Vg}^0, since the equilibrium volume of a solid changes with T. C_V can be converted to C_{Vg}^0 by subtracting the anharmonic contribution which is also approximately linear with respect to temperature [10,11]:

$$C_V = a_1 T + C_V^0. \tag{7}$$

Combining eq. (6) and eq. (7) gives

$$C_{pg} = aT + C_{Vg}^0, \tag{8}$$

where a $(= a_1 + a_2)$ is a constant.

The value of the parameter a can be estimated. a_2 can be calculated using the thermodynamic data. Values for V_m, α_V, and K_T at room temperatures for NBS-710 glass are [12]: $V_m = 2.44 \times 10^{-5}$ m^3 mol^{-1}, $\alpha_V = 3 \times 10^{-5}$ K^{-1}, and $K_T^{-1} = 4.26 \times 10^{10}$ Pa. These give a value of 0.13 $\times 10^{-4}$ K^{-1} for $a_2/3R$. The value of $a_1/3R$ has been reported to be in the range of 0.4×10^{-4} K^{-1} to 1.7×10^{-4} K^{-1} [10,11,13]. Thus, the value of $a/3R$ should lie in the range 0.5×10^{-4} K^{-1} to 1.8×10^{-4} K^{-1}. In subsequent analysis, we assume $a/3R$ as the mid-point of this range $(= 1.2 \times 10^{-4}$ K$^{-1})$ for the NBS-710 glass.

2.2.1. The quasi-harmonic model

The temperature dependence of the heat capacity of crystals has been examined by Born, von Karman, Blackman, Montroll etc. (see for example ref. [14]) on the basis of lattice dynamics. In the quasi-harmonic approximation, the crystal vibrations are assumed to be harmonic but are allowed to depend on the volume [15,16]. According to the quasi-harmonic model, C_{pg} is expressed as

$$C_{pg} = aT + 9R\left[\frac{T}{\theta_{(T)}}\right]^3 \int_0^{\theta_{(T)}/T} \frac{x^4 e^x}{(e^x - 1)^2} dx, \tag{9}$$

where, according to Domb and Salter [8], for $T > \theta_\infty/2\pi$, $\theta_{(T)}$ is given by

$$\theta_{(T)}^2 = \theta_\infty^2\{1 - \mu[\theta_\infty/T]^2 + \mu'[\theta_\infty/T]^4 - \dots\}. \tag{10}$$

Here, θ_∞, μ and μ' are constants. The value of θ_∞ is, in general, larger than the value of the Debye temperature which is obtained from either low temperature heat capacity or from elastic constants. μ and μ' are characteristic functions of the phonon density of states, $g(\nu)$. For example, the parameter μ is defined as [9]

$$\mu = \frac{3}{100}\left[\frac{\overline{\nu^4}}{\left(\overline{\nu^2}\right)^2} - \frac{25}{21}\right],$$

where

$$\overline{\nu^n} \equiv \frac{1}{3N}\int_0^\infty \nu^n g(\nu)\, d\nu.$$

If terms of order higher than the second are neglected in eq. (10) (as is assumed in this paper), eq. (9) involves three fitting parameters: a, θ_∞ and μ. The quasi-harmonic model involves only two parameters when $a/3R$ is known. It should be noted that when $\mu = 0$, the quasi-harmonic model reduces to the well known Debye model.

2.2.2. The three band model

The three band model, proposed by Hiro et al. [9] for oxide glasses, considers C_{Vg}^0 to be composed of separate contributions from the network formers (C_1) and from the network modifiers (C_2). The contribution from the network modifiers is described by the Einstein model with θ_E as a characteristic temperature. C_1 is postulated as a sum of the heat capacities of a three-dimensional continuum for low frequency model (from 0 to θ_3) and of a one-dimensional continuum corresponding to high frequency modes (from θ_3 to θ_1). Thus, C_{Vg}^0 is given by the following equations:

$$C_{Vg}^0 = M_1 C_1 + M_2 C_2, \tag{11}$$

where

$$C_1 = 3R\left(\frac{T}{\theta_1}\right)\int_{\theta_3/T}^{\theta_1/T} \frac{x^2 e^x}{(e^x - 1)^2} dx + 9R\left(\frac{T}{\theta_3}\right)^3\left(\frac{\theta_3}{\theta_1}\right)\int_0^{\theta_3/T} \frac{x^4 e^x}{(e^x - 1)^2} dx;$$

$$C_2 = 3R\frac{y^2 e^y}{(e^y - 1)^2}, \quad \text{where } y = \theta_E/T.$$

In eq. (11), M_1 is the number of g-atms of network-forming ions, and M_2 is the number of g-atms of network-modifying ions per formula unit. For example, $M_1 = 0.5 + 1.5$ and $M_2 = 0.5$ for $Mg_{0.5}Si_{0.5}O_{1.5}$ glass. Equation (11) involves four fitting parameters (a, θ_1, θ_3 and θ_E) in addition to the knowledge of M_1 and M_2. However, it reduces to a three parameter model when the value of a is known.

3. Experimental procedure

The NBS-710 glass, a viscosity standard issued by the National Institute of Standards and Technology, was chosen for the present investigation because of its stability with respect to crystallization, chemical attack and phase separation. Its composition is: SiO_2 70.5 wt%, Na_2O 8.7 wt%, K_2O 7.7 wt%, CaO 11.6 wt%, Sb_2O_3 1.1 wt%, SO_3 0.2 wt%, and R_2O_3 0.2 wt%. After annealing at 973 K ($T_g + 100$ K) for 2 h, samples were cooled at constant rates. Four samples B1, B2, B3 and B4 were prepared with cooling rates of 2, 5, 20 and 40 K min^{-1} respectively. Heat capacities of these samples were measured on a Perkin-Elmer DSC 7 Differential Scanning Calorimeter using a heating rate of 40 K min^{-1} from 333 to 973 K. To calculate the absolute values of heat capacity, before each sample scan, two other DSC scans were performed: one with an empty sample pan and another using sapphire which is a heat capacity standard reference material [17].

4. Results

Figure 1 shows the heat capacity results for the four samples. The heat capacity values for all four samples agree up to 650 K. A sample annealed at 650 K for 15 days also gave values of C_p in agreement with the rate cooled samples for $T < 650$ K. This indicates that below 650 K, there is no enthalpy relaxation in times less than 15 days and that C_{pg} is independent of the thermal history. The heat capacity behaviors of these sam-

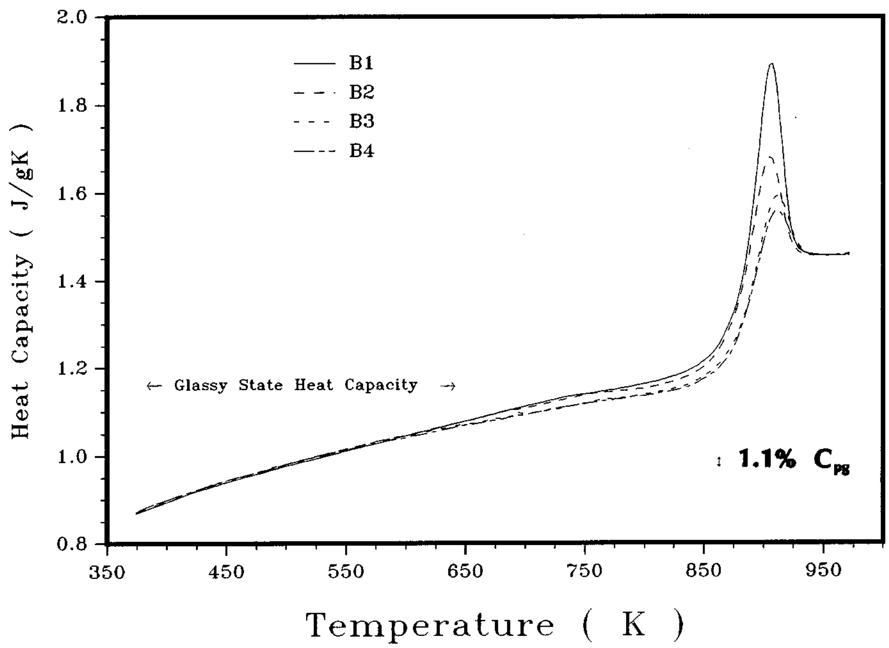

Fig. 1. Measured heat capacity values of the four samples vs. temperature obtained by heating at a rate of 40 K min^{-1} in the DSC. Differences between samples in the temperature range from 650 to 950 K are due to differences in their thermal history dependent relaxation. Since no significant relaxation occurs in experimental time scale for $T < 650$ K, C_p in this range is considered to be equal to C_{pg}.

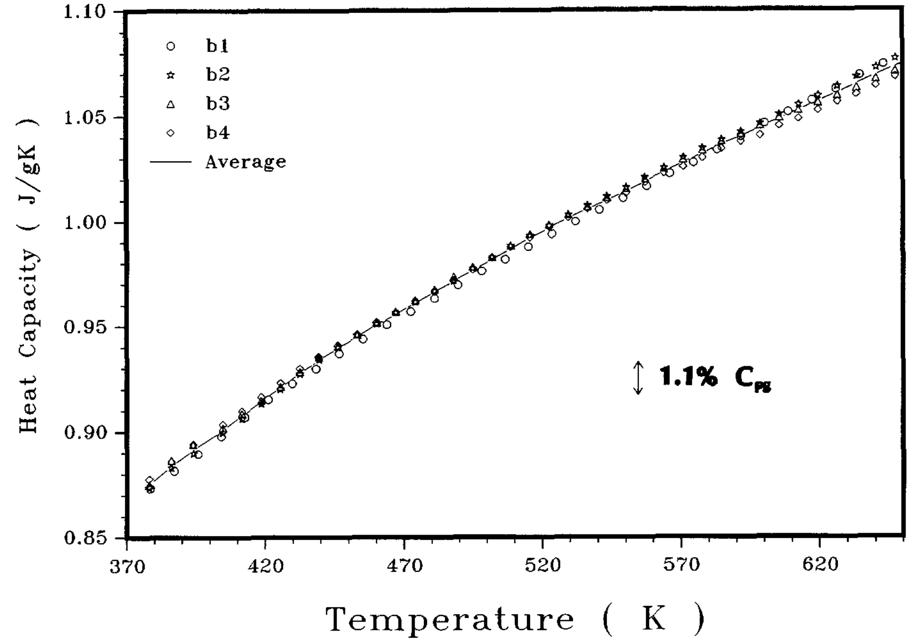

Fig. 2. Average glassy state heat capacity vs. temperature in the temperature range of 370–650 K for NBS-710 glass. The points represent the four samples. The curve gives the averaged behavior.

ples differ in the temperature range from 650 to 950 K due to structural relaxation which is influenced by their thermal histories. In the liquid state ($T > 950$ K), the heat capacities of all samples approach the same (temperature-independent) value of 1.457 J g^{-1} K^{-1}. Figure 2 shows the average values of C_{pg} in the temperature range from 370 to 650 K. The uncertainty in the C_p measurement is estimated to be 1.1%.

5. Discussion

The average values shown in fig. 2 were fit to the seven heat capacity expressions (eqs. (1)–(5), (9) and (11)). Figure 3 shows the best fits for the five empirical expressions: linear expression (eq. (3)); Scherer expressions (eqs. (1) and (2)); the Maier–Kelly expression (eq. (4)); and the Tsao expression (eq. (5) with $n = 3$). As can been seen from this figure, the Tsao expression and the Maier–Kelly expression describe the data well. However, the linear expression and the Scherer expression (eq. (1)) fail as indicated by the systematic nature of deviations, and the fact that the deviations exceed the experimental uncertainty at some temperatures. For the Scherer expression (eq. (2)), the fit is acceptable. However, the value of A obtained from the fit is 1.358 J g^{-1} K^{-1}, which differs significantly from the measured C_{pl} of 1.457 J g^{-1} K^{-1}. Therefore, eq. (2) is considered unacceptable.

Figure 4 shows the best fits for the two theoretical models: the three band model, and the quasi-harmonic model. Both models provide excellent fits to the data. A simple Debye model (corresponding to $\theta_{(T)} =$ constant in eq. (9)) gave a poor fit indicating the inadequacy of the Debye model to describe the heat capacity of the NBS-710 glass.

The best fit parameters of all models are listed in table 1. Also listed are the residual mean squares associated with these fits. Based on the residual mean square values only, the quasi-harmonic model, the three band model, the Maier–Kelly expression, and the Tsao expression, describe the data well.

Clearly, the Maier–Kelly expression and the

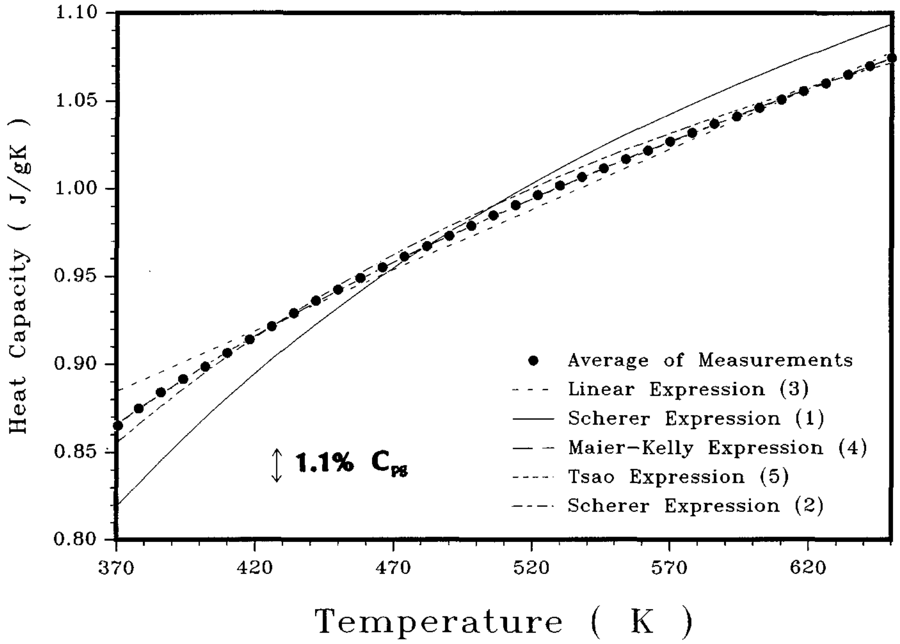

Fig. 3. Fits of five empirical heat capacity models: linear, Scherer and the Maier–Kelly expression, and the Tsao expression. Both the Maier–Kelly expression and the Tsao expression provide excellent fits. Fits of the linear expression and the two Scherer expressions are poor.

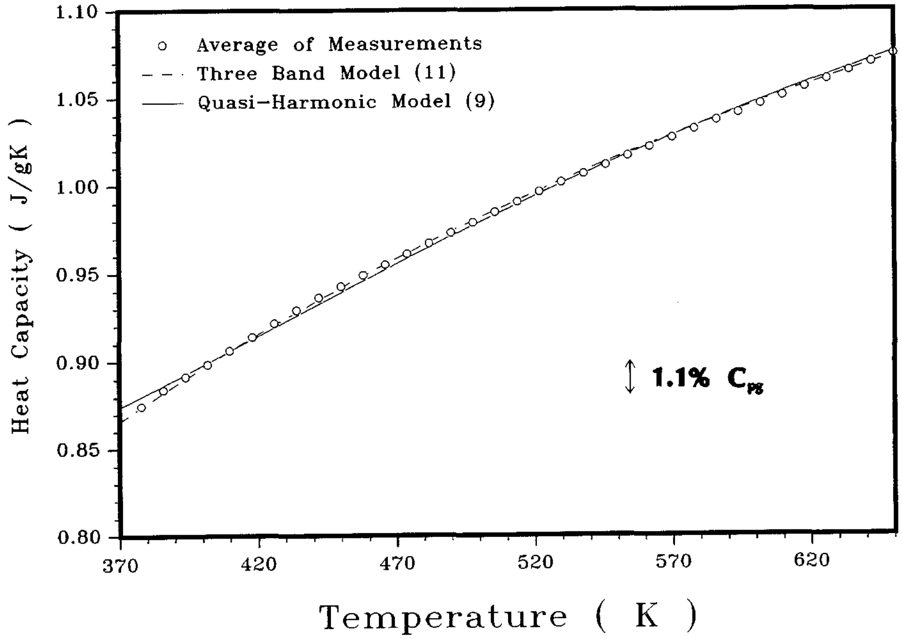

Fig. 4. Fits of two theoretical heat capacity models: the three band model and the quasi-harmonic model. Both models fit equally well.

Table 1
Values of best fitting parameters

Model	Best fitting parameters		RMS [a)] $J^2\ g^{-2}\ K^{-2}$
$C_{pg} = C_{pl}^{(b)} - (B/T)$	$B = 236.1$	$J\ g^{-1}$	3.72×10^{-4}
$C_{pg} = A + BT$	$A = 0.629$	$J\ g^{-1}\ K^{-1}$	4.72×10^{-5}
	$B = 6.906 \times 10^{-4}$	$J\ g^{-1}\ K^{-2}$	
$C_{pg} = A - (B/T)$	$A = 1.358$	$J\ g^{-1}\ K^{-1}$	3.44×10^{-5}
	$B = 186.0$	$J\ g^{-1}$	
$C_{pg} = \dfrac{T^3(A+BT)}{T^3 + T_0^3}$	$A = 0.784$	$J\ g^{-1}\ K^{-1}$	1.06×10^{-5}
	$B = 4.805 \times 10^{-4}$	$J\ g^{-1}\ K^{-2}$	
	$T_0 = 178.0$	K	
$C_{pg} = A + BT - (C/T^2)$	$A = 0.828$	$J\ g^{-1}\ K^{-1}$	1.06×10^{-5}
	$C = 4.418 \times 10^{-4}$	$J\ g^{-1}\ K^{-2}$	
	$C = 17185.9$	$J K\ g^{-1}$	
Three band model ($a/3R = 1.2 \times 10^{-4}\ K^{-1}$)	$\theta_1 = 1106.0$	K	1.25×10^{-5}
	$\theta_2 = 68.9$	K	
	$\theta_E = 2283.8$	K	
Quasi-harmonic model ($a/3R = 1.2 \times 10^{-4}\ K^{-1}$)	$\theta_\infty = 1256.9$	K	1.55×10^{-5}
	$\mu = 0.03$		

[a)] Residual mean square $\equiv \Sigma_{i=1}^{N}(y_i - Y_i)^2/(N-P)$; P = number of fitting parameters.
[b)] C_{pl} is found to be 1.457 J g^{-1} K^{-1} for NBS-710 glass.

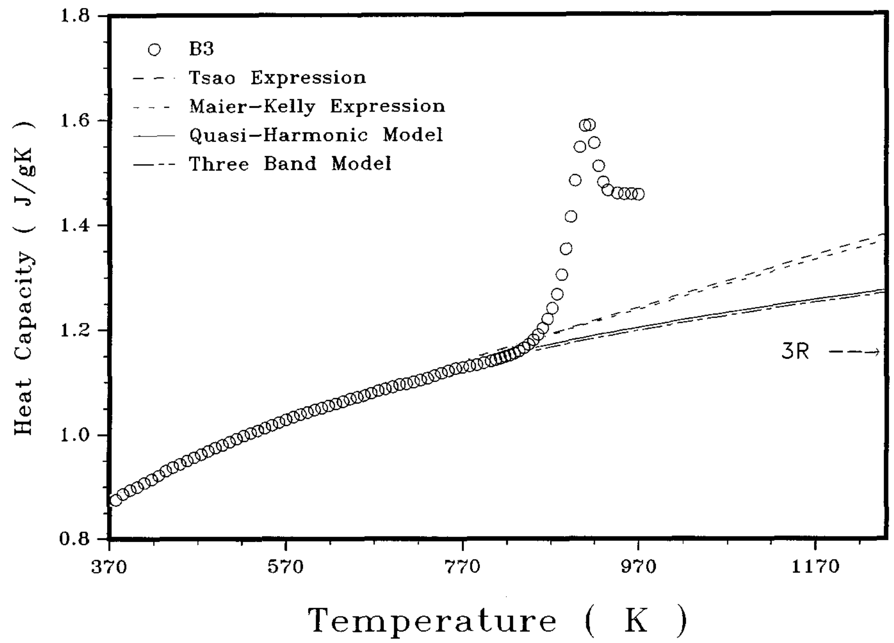

Fig. 5. Extrapolations based on the quasi-harmonic model, the three band model, the Maier–Kelly expression, and the Tsao expression for $T > 650$ K. They differ significantly in the high temperature range. Also shown by the arrow is the Dulong–Petit limit (1.156 J g^{-1} K^{-1}).

Tsao expression are simpler than the other two. However, they are strictly empirical. Interestingly, their extrapolations lie much above those based on the two theoretical models at high temperatures (see fig. 5). Thus, to choose a model for the extrapolation of $C_{pg}(T)$ to $T > T_g$ requires evaluation of the values of the fitting parameters of these models.

In the Maier–Kelly and Tsao expressions, the term A is typically close to the Dulong–Petit value [7]. For the Maier–Kelly expression, the best value of A (0.828 J g^{-1} K^{-1}) is 28% smaller than the Dulong–Petit value (1.156 J g^{-1} K^{-1}). For the Tsao expression, term A (0.784 J g^{-1} K^{-1}) is 32% less than the Dulong–Petit value. This lack of agreement between the values of A and the Dulong–Petit limit leaves the validities of these models open to question. The fact that both theoretical models give similar values of $C_p(T)$ at $T > T_g$ gives confidence in extrapolation based on them.

In the three band model, the value of θ_1 is related to the Debye temperature for the Si–O bond, which has a value of ~ 1300 K [9]. This is close to the present value of 1106 K obtained for NSB-710 glass. The values of θ_3 and θ_E should be considerably less than the value of θ_1 [9]. For the present fit, the value of 2284 K for θ_E far exceeds θ_1. Varying the value of a in eq. (8) did not decrease θ_E significantly. This leaves the application of the three band theory to present heat capacity data questionable possibly because $C_p(T)$ measurements have not been made at low enough temperatures.

In the quasi-harmonic model, the values of μ have been published only for few materials. For example, Barron et al. [15] have reported μ of: 0.02 for NaI; 0.003 for NaCl; 0.012 for KI. These values of μ are close to the present value of 0.03 obtained for the NBS-710 glass.

The values of the parameter θ_∞ are not available for oxide materials. To ascertain the reasonableness of the present θ_∞ value, the quasi-harmonic model was applied to the published $C_p(T)$ data for alumina [17], quartz [18], and to the $C_{pg}(T)$ data for SiO_2-glass [18]. The computed values of θ_∞ are: 1081 K for Al_2O_3, 1322 K for quartz; and 1022 K for SiO_2-glass. The range covered by these θ_∞ values includes the value (1257 K) obtained for the NBS-710 glass. Thus, the values of the fitting parameters for the quasi-harmonic model appear to be reasonable. This, coupled with the fact that the quasi-harmonic model has less number of fitting parameters than the three band model, leads us to choose the quasi-harmonic model as the preferred model.

6. Conclusions

The quasi-harmonic model, the three band model, the Maier–Kelly expression, and the Tsao expression adequately describe the glassy state heat capacity of the glass studied in the temperature range of 370–650 K (within experimental uncertainty). The other models (linear expression, Scherer's expressions) fail to give a satisfactory description. Maier–Kelly and Tsao expressions give values of C_p for $T > T_g$ similar but significantly higher than the two theoretical models. Between the two theoretical models, the quasi-harmonic model is preferred because it has less number of fitting parameters and the values of the fitting parameters are reasonable.

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